

|                                                        |                                                     |                                      |                                       |                                                |
|--------------------------------------------------------|-----------------------------------------------------|--------------------------------------|---------------------------------------|------------------------------------------------|
| This is sample data provided for CALIBER purposes only |                                                     | ONE POINT LESSON (OPL)               |                                       | OPL No: <b>OPL-GA-1201A-06</b><br>Sheet 1 of 1 |
| OPL Title                                              | Start-Up & Priming Procedure GA-1201A               | Discipline                           | Process / Operations                  |                                                |
| Equipment                                              | GA-1201A - HEXANE FEED PUMP                         | Area / Unit                          | 1200 - HEXANE FEED SYSTEM             |                                                |
| Related Interlock                                      | SEQ-1201 (SIL 1)                                    | P&ID; Ref                            | TJC-LLD-PID-1201                      |                                                |
| Classification:                                        | <input checked="" type="checkbox"/> Basic Knowledge | <input type="checkbox"/> Improvement | <input type="checkbox"/> Trouble Case | Aspect: P-Q-C-D-S-M-E                          |

### 1. PURPOSE / OBJECTIVE

This One Point Lesson explains **start-up & priming procedure ga-1201a** for GA-1201A (HEXANE FEED PUMP) in the HEXANE FEED SYSTEM. Correct priming and venting prevents a dry-run that instantly destroys the mechanical seal. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

### 2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **n-Hexane** at up to 16 barg / 80 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

### 3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

### 4. DETAILED PROCEDURE / STEPS

| Step | Action                                                                        | Check / Acceptance         |
|------|-------------------------------------------------------------------------------|----------------------------|
| 1    | Open the suction valve fully and vent the casing air via N5 vent              | Permit valid & briefed     |
| 2    | Establish seal flush Plan 11 and confirm PDI-1201 dP                          | Parameter within limit     |
| 3    | Confirm the min-flow recycle line FV-1201 is open                             | Interlock status confirmed |
| 4    | Bump-start the motor and check correct rotation direction                     | Change logged on DCS       |
| 5    | Start the motor and confirm PIC-1201 pressure and vibration settle within 5 s | Handover to next shift     |

### 5. COMMON PROBLEMS & TROUBLESHOOTING

| Symptom                                                               | Likely Cause                                                          | Action                                                                |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| Hexane leak observed at GA-1201A seal gland during operation, seal dr | API Plan 11 flush orifice RO-1201 partially plugged causing seal face | Replaced JC T2100 mechanical seal cartridge, cleaned flush orifice, v |
| GA-1201A tripped on VSHH-1201 high vibration 7.4 mm/s                 | Angular misalignment 0.12 mm/100mm after foundation settlement        | Laser re-aligned pump/motor, re-shimmed motor feet, vibration back to |
| Bearing DE noisy with rising temperature TI-1201 trend                | Outer race spalling on DE bearing due to prolonged misalignment       | Renewed DE bearing 7310 BECBM, flushed housing, refilled ISO VG 68    |

### 6. KEY LEARNING POINTS

- Correct priming and venting prevents a dry-run that instantly destroys the mechanical seal.
- Respect the interlock SEQ-1201: e.g. PSL-1201 trips at < 0.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-1201 before working on GA-1201A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

| Prepared by                 | Reviewed by (Supervisor) | Approved by (Manager)    | Date of Sharing         |
|-----------------------------|--------------------------|--------------------------|-------------------------|
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